**UNIVERSIDAD DE LAS FUERZAS ARMADAS ESPE**

Department of Computer Science

Software Engineering

Advanced Web Development

Hyper-scale Technical Management Limited - Team

# API RESTful URI Design Document

## System Endpoints & Resource Architecture

**Group Members:** Galarza César, Gualotuña Brayan, Erazo Isaac

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**Project:** Advanced Web System for "Ser salud" Comprehensive Clinic

**Customer:** Flores Villalva David Alejandro

*"What is important is not being able, what is important is trying*

# 1. Introduction to the API Architecture

This document outlines the **Uniform Resource Identifier (URI)** design and the **RESTful endpoints collection** for the "Ser salud" Clinical Management System. The backend architecture has been engineered using Node.js and Express, exposing a secure, stateless **API** that facilitates communication between the client interfaces and the **PostgreSQL database.**

The purpose of this document is to formally present the 24 specific endpoints required to satisfy the system's operational needs, detailing their accessibility, HTTP methods, and strict adherence to REST architectural constraints.

# 2. URI Design Principles Applied

To guarantee a scalable and predictable integration with the frontend, the following **RESTful design** principles were strictly applied in the **construction of the URIs:**

- **Nouns Over Verbs: URIs** are constructed using nouns representing the domain entities rather than actions. The action is uniquely defined by the **HTTP method** (GET, POST, PUT, DELETE).
- **Pluralization:** All core resource **URIs** are pluralized to indicate collections of resources, **/api/medicos** instead of **/api/medico**, maintaining semantic consistency.
- **Hierarchical Routing:** Relationships and specific queries are structured hierarchically using path variables and query parameters, **/api/medicos/especialidad/:esp** to filter medical staff by branch, or **\fecha=YYYY-MM-DD** for chronological filtering.
- **Stateless Authentication:** Secure routes mandate a **JWT Bearer token** passed in the \texttt{Authorization} header, adhering to stateless session management.

# 3. Base Configuration

All endpoints listed in the subsequent catalog are appended to the core **API environment route.**

1. **Development Base URL:** http://localhost:3000/api
2. **Production Base URL:** To be defined upon ***Vercel/Render*** **deployment**
3. **Authentication Format:** Authorization = Bearer **<token>**

# 4. Complete Endpoint Catalog (URI Collection)

The following table consolidates the complete collection of 24 URIs developed for the clinic's operational scope, categorized by their corresponding module and authorization level.

| # | Method | URI (Endpoint) | Auth Reqd. | Role |
|---|---|---|---|---|
| **Authentication & System Health** | | | | |
| 1 | GET | /health | No | — |
| 2 | POST | /auth/register | No | — |
| 3 | POST | /auth/login | No | — |
| 4 | GET | /auth/me | Yes | All |
| 5 | POST | /auth/logout | Yes | All |
| **Therapies Management** | | | | |
| 6 | GET | /terapias | No | — |
| 7 | GET | /terapias?especialidad=X | No | — |
| 8 | GET | /terapias/:id | No | — |
| 9 | POST | /terapias | Yes | Admin |
| 10 | PUT | /terapias/:id | Yes | Admin |
| 11 | DELETE | /terapias/:id | Yes | Admin |
| **Medical Staff** | | | | |
| 12 | GET | /medicos | No | — |
| 13 | GET | /medicos/:id | No | — |
| 14 | GET | /medicos/especialidad/:esp | No | — |
| **Appointments & Scheduling** | | | | |
| 15 | GET | /citas/horarios-disponibles | Yes | All |
| 16 | POST | /citas | Yes | Patient |
| 17 | GET | /citas | Yes | All |
| 18 | GET | /citas/:id | Yes | All |
| 19 | PUT | /citas/:id | Yes | Specialist / Admin |
| 20 | DELETE | /citas/:id | Yes | Patient |
| **User Profiles** | | | | |
| 21 | GET | /users | Yes | Admin |
| 22 | GET | /users/:id | Yes | User / Admin |

| | | | | |
|---|---|---|---|---|
| 23 | PUT | /users/:id | Yes | User / Admin |
| 24 | DELETE | /users/:id | Yes | Admin |

# 5. Authentication, Security & Session Management

The perimeter security and access control of the “Ser salud” platform are governed by a stateless authentication architecture based on **JSON Web Tokens (JWT).** This section details the endpoints responsible for managing the user identity lifecycle, from initial registration with **encrypted credentials (hash)** to session invalidation. The design of these endpoints ensures that passwords are never transmitted or returned in plain text, and that every subsequent request to the clinical modules is protected by an authorization token in the **HTTP headers.** A **“Health Check” endpoint** is included, which is essential for automated infrastructure monitoring and load balancing, ensuring that the microservice is operational before processing sensitive transactions.

## *1. Health Check*

**GET** {{base_url}}/api/health

**Without a body**

## *2. Register User (Patient)*

**POST** {{base_url}}/api/auth/register

**Content-Type:** application/json

**Body:**

**json**

```
{
    "cedula": "1726543210",
    "fullName": "Pedro Sánchez Morales",
    "email": "pedro.sanchez@email.com",
    "telefono": "0987654325",
    "tipoSeguro": "iess",
    "password": "password123"
}
```

**Expected response:**

**json**

{

"message": "Usuario registrado exitosamente",

"user": {

"id": 9,

"cedula": "1726543210",

"fullName": "Pedro Sánchez Morales",

"email": "pedro.sanchez@email.com",

"telefono": "0987654325",

"tipoSeguro": "iess",

"role": "paciente",

"createdAt": "2026-05-02T04:00:00.000Z"

}

}

## 3. *Login*

**POST** {{base_url}}/api/auth/login

**Content-Type:** application/json

**Body (Patient):**

**json**

{

"cedula": "1234567890",

"password": "password123"

}

**Body (Admin):**

**json**

```json
{
    "cedula": "admin",
    "password": "admin123"
}
```

**Body (Doctor):**

**json**

```json
{
    "cedula": "1111111111",
    "password": "password123"
}
```

**Expected response:**

**json**

```json
{
    "message": "Login exitoso",
    "token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJ1c2VySWQiOjIsImNlZHVsYSI6IjEyMzQ1Njc4OTAiLCJyb2xlIjoicGFjaWVudGUiLCJmdWxsTmFtZSI6Ikp1YW4gUMOpcmV6IEdhcmPDrWEiLCJpYXQiOjE3MTQ2MjQ4MDAsImV4cCI6MTcxNTIyOTYwMH0.xxxxx",
    "user": {
        "id": 2,
        "cedula": "1234567890",
        "username": "1234567890",
        "fullName": "Juan Pérez García",
        "email": "juan.perez@email.com",
        "telefono": "0987654321",
        "tipoSeguro": "iess",
        "role": "paciente"
```

}

}

**IMPORTANT:** Save the `token` to use it with the following endpoints.

### *4. Get Current User*

**GET** {{base_url}}/api/auth/me

**Authorization:** Bearer {{token_paciente}}

**Without a body**

**Expected response:**

**json**

{

"id": 2,

"cedula": "1234567890",

"username": "1234567890",

"fullName": "Juan Pérez García",

"email": "juan.perez@email.com",

"telefono": "0987654321",

"tipoSeguro": "iess",

"role": "paciente",

"createdAt": "2026-05-02T03:00:00.000Z",

"updatedAt": "2026-05-02T03:00:00.000Z",

"medico": null

}

### *5. Log Out*

**POST** {{base_url}}/api/auth/logout

**Authorization:** Bearer {{token_paciente}}

**Without a body**

**Expected response:**

**json**

{

"message": "Sesión cerrada exitosamente",

"timestamp": "2026-05-04T04:35:00.000Z"

}

**Note:** In a stateless JWT architecture, logout is primarily handled on the front end by removing the token from local storage.

This endpoint **is used to** log logout audit entries, invalidate tokens if a blacklist is implemented in the future, and clear session data from the server if it exists.

**After logout**, the token must be removed from local storage on the frontend, the user must be redirected to the login page, and that token must not be used for future requests.

## 6. Authentication, Security & Session Management

The transactional core of the system lies in the Therapies and Medical Appointments modules, which expose highly structured endpoints to support the clinic's business logic. The **URIs designed** in this section enable the manipulation of complex resources, managing interconnected relationships between patients, specialists, and treatments. To ensure the integrity of clinical data, the **JSON payloads** sent to these endpoints undergo rigorous validation on the backend, such as a 24-hour advance scheduling restriction, appointment scheduling conflict prevention **(anti-overlap)**, and the parametric assignment of medical records and attached test results. The responses from these endpoints are formatted to provide the client with complete nested objects, reducing the need for multiple **API calls** and optimizing the interface's rendering time.

**Nota para su compañero (Brayan - Lead Developer):** Él debe encargarse de esta sección para que copie y pegue aquí la sección **"💊 TERAPIAS" (Endpoints del 6 al 11)** y la sección **"📅 CITAS" (Endpoints del 15 al 20)**.

### *6. View All Therapies*

### Obtener Todas las Terapias

GET {{base_url}}/api/terapias

Sin body

Respuesta esperada:

```json

[
  {
    "id": 1,
    "nombre": "Fisioterapia Deportiva",
    "descripcion": "Tratamiento especializado para lesiones deportivas y recuperación de atletas. Incluye técnicas de rehabilitación avanzadas.",
    "especialidad": "Fisioterapia",
    "duracion": 60,
    "precio": "45.00",
    "imagen": "https://picsum.photos/seed/fisio1/400/300",
    "activa": true,
    "createdAt": "2026-05-02T03:43:01.000Z",
    "updatedAt": "2026-05-02T03:43:01.000Z"
  }
]

**Obtener Terapias por Especialidad**

GET {{base_url}}/api/terapias?especialidad=Fisioterapia

Sin body

[
    {
        "id": 13,
```

"nombre": "Fisioterapia Deportiva",

"descripcion": "Tratamiento especializado para lesiones deportivas y recuperación de atletas. Incluye técnicas de rehabilitación avanzadas.",

"especialidad": "Fisioterapia",

"duracion": 60,

"precio": "45",

"imagen": "https://picsum.photos/seed/fisio1/400/300",

"activa": true,

"createdAt": "2026-05-04T15:44:25.159Z",

"updatedAt": "2026-05-04T15:44:25.159Z"

},

{

"id": 14,

"nombre": "Rehabilitación Post-Quirúrgica",

"descripcion": "Programa de rehabilitación integral después de cirugías ortopédicas. Recuperación funcional completa.",

"especialidad": "Fisioterapia",

"duracion": 75,

"precio": "65",

"imagen": "https://picsum.photos/seed/fisio2/400/300",

"activa": true,

"createdAt": "2026-05-04T15:44:25.469Z",

"updatedAt": "2026-05-04T15:44:25.469Z"

}

]

**Crear Terapia**

POST {{base_url}}/api/terapias

Authorization: Bearer {{token_admin}}

Content-Type: application/json

Body:

```json

{

"nombre": "Masaje Terapéutico",

"descripcion": "Masaje especializado para aliviar tensiones musculares y mejorar la circulación sanguínea",

"especialidad": "Fisioterapia",

"duracion": 45,

"precio": 40.00,

"imagen": "https://picsum.photos/seed/masaje/400/300"

}

**Respuesta esperada:**

```json

{

"message": "Terapia creada exitosamente",

"terapia": {

"id": 7,

"nombre": "Masaje Terapéutico",

"descripcion": "Masaje especializado para aliviar tensiones musculares y mejorar la circulación sanguínea",

"especialidad": "Fisioterapia",

"duracion": 45,

"precio": "40.00",

"imagen": "https://picsum.photos/seed/masaje/400/300",

"activa": true,

"createdAt": "2026-05-02T04:00:00.000Z",

"updatedAt": "2026-05-02T04:00:00.000Z"

}

}

**Actualizar Terapia**

PUT {{base_url}}/api/terapias/1

Authorization: Bearer {{token_admin}}

Content-Type: application/json

Body:

json

{

"nombre": "Fisioterapia Deportiva Avanzada",

"precio": 50.00,

"activa": true

}

Respuesta esperada:

```json
{
  "message": "Terapia actualizada exitosamente",
  "terapia": {
    "id": 1,
    "nombre": "Fisioterapia Deportiva Avanzada",
    "descripcion": "Tratamiento especializado para lesiones deportivas...",
    "especialidad": "Fisioterapia",
    "duracion": 60,
    "precio": "50.00",
    "imagen": "https://picsum.photos/seed/fisio1/400/300",
    "activa": true,
    "createdAt": "2026-05-02T03:43:01.000Z",
    "updatedAt": "2026-05-02T04:05:00.000Z"
  }
}
```

**Eliminar Terapia (Admin)**

DELETE {{base_url}}/api/terapias/7

Authorization: Bearer {{token_admin}}

Sin body

Respuesta esperada:

{

"message": "Terapia eliminada exitosamente"

}

**Obtener Horarios Disponibles**

GET
{{base_url}}/api/citas/horarios-disponibles?medicoId=1&fecha=2026-05-10

Authorization: Bearer {{token_paciente}}

Sin body

Query params:

- `medicoId`: ID del médico (requerido)

- `fecha`: Fecha en formato YYYY-MM-DD (requerido)

Respuesta esperada:

[
  {
    "fecha": "2026-05-10",
    "hora": "08:00",
    "disponible": true,
    "medicoId": 1
  },
  {
    "fecha": "2026-05-10",
    "hora": "08:30",
    "disponible": true,
    "medicoId": 1
  },

{

"fecha": "2026-05-10",

"hora": "09:00",

"disponible": false,

"medicoId": 1

}

]

**Crear Cita (Paciente)**

POST {{base_url}}/api/citas

Authorization: Bearer {{token_paciente}}

Content-Type: application/json

Body:

{

"medicoId": 1,

"terapiaId": 1,

"fecha": "2026-05-10",

"hora": "10:00",

"sintomas": "Dolor en rodilla derecha después de correr. Necesito evaluación para continuar con mi entrenamiento deportivo.",

"tieneExamenes": false

}

Body con exámenes:

```json
```

{

"medicoId": 1,

"terapiaId": 1,

"fecha": "2026-05-10",

"hora": "14:00",

"sintomas": "Dolor crónico en espalda baja. Tengo radiografías recientes.",

"tieneExamenes": true,

"examenes": ["radiografia_lumbar.pdf", "resonancia_magnetica.pdf"]

}

Respuesta esperada:

{

"message": "Cita creada exitosamente",

"cita": {

"id": 4,

"fecha": "2026-05-10",

"hora": "10:00",

"estado": "pendiente",

"paciente": {

"fullName": "Juan Pérez García",

"cedula": "1234567890",

"email": "juan.perez@email.com"

},

"medico": {

"fullName": "Dr. Carlos Mendoza Silva",

      "especialidad": "Fisioterapia"
    },
    "terapia": {
      "nombre": "Fisioterapia Deportiva",
      "duracion": 60,
      "precio": 45
    }
  }
}


**Obtener Mis Citas**

GET {{base_url}}/api/citas

Authorization: Bearer {{token_paciente}}

Sin body

Query params opcionales:

- `estado`: Filtrar por estado (pendiente, confirmada, completada, cancelada)

- `fecha`: Filtrar por fecha (YYYY-MM-DD)

Ejemplos:

GET {{base_url}}/api/citas?estado=pendiente

GET {{base_url}}/api/citas?fecha=2026-05-10

Respuesta esperada:

```json
[
  {
    "id": 1,
```

"pacienteId": 2,

"medicoId": 1,

"terapiaId": 1,

"fecha": "2026-05-05",

"hora": "10:00",

"estado": "pendiente",

"sintomas": "Dolor en rodilla derecha después de correr. Necesito evaluación para continuar entrenamiento.",

"tieneExamenes": false,

"examenes": [],

"motivoCancelacion": null,

"notasMedico": null,

"createdAt": "2026-05-02T03:43:01.000Z",

"updatedAt": "2026-05-02T03:43:01.000Z",

"paciente": {

"id": 2,

"fullName": "Juan Pérez García",

"cedula": "1234567890",

"email": "juan.perez@email.com",

"telefono": "0987654321",

"tipoSeguro": "iess"

},

"medico": {

"id": 1,

```
      "fullName": "Dr. Carlos Mendoza Silva",
      "especialidad": "Fisioterapia",
      "numeroLicencia": "MED-FIS-001",
      "calificacion": 4.8
    },
    "terapia": {
      "id": 1,
      "nombre": "Fisioterapia Deportiva",
      "descripcion": "Tratamiento especializado para lesiones deportivas...",
      "especialidad": "Fisioterapia",
      "duracion": 60,
      "precio": 45
    }
  }
]
```


**Obtener Cita por ID**

GET {{base_url}}/api/citas/1

Authorization: Bearer {{token_paciente}}

Sin body

**Actualizar Cita (Médico/Admin)**

PUT {{base_url}}/api/citas/1

Authorization: Bearer {{token_medico}}

Content-Type: application/json

Body (Médico agrega notas):

```json

{

"notasMedico": "Paciente presenta mejoría significativa en la movilidad de la rodilla. Se recomienda continuar con ejercicios de fortalecimiento en casa y regresar en 2 semanas para evaluación."

}

Body (Cambiar estado):

```json

{

"estado": "completada",

"notasMedico": "Sesión completada exitosamente. Paciente responde bien al tratamiento."

}

Respuesta esperada:

```json

{

"message": "Cita actualizada exitosamente",

"cita": {

"id": 1,

"pacienteId": 2,

"medicoId": 1,

"terapiaId": 1,

"fecha": "2026-05-05T00:00:00.000Z",

"hora": "10:00",

"estado": "completada",
```

"sintomas": "Dolor en rodilla derecha...",

"tieneExamenes": false,

"examenes": [],

"motivoCancelacion": null,

"notasMedico": "Sesión completada exitosamente. Paciente responde bien al tratamiento.",

"createdAt": "2026-05-02T03:43:01.000Z",

"updatedAt": "2026-05-02T04:10:00.000Z"

}

}

**Cancelar Cita (Paciente)**

DELETE {{base_url}}/api/citas/1

Authorization: Bearer {{token_paciente}}

Content-Type: application/json

Body:

```json

{

"motivo": "Tengo un compromiso urgente de trabajo que no puedo posponer"

}

Respuesta esperada:

```json

{

"message": "Cita cancelada exitosamente",

"cita": {

"id": 1,

"pacienteId": 2,

"medicoId": 1,

"terapiaId": 1,

"fecha": "2026-05-05T00:00:00.000Z",

"hora": "10:00",

"estado": "cancelada",

"sintomas": "Dolor en rodilla derecha...",

"tieneExamenes": false,

"examenes": [],

"motivoCancelacion": "Tengo un compromiso urgente de trabajo que no puedo posponer",

"notasMedico": null,

"createdAt": "2026-05-02T03:43:01.000Z",

"updatedAt": "2026-05-02T04:15:00.000Z"

}

}

## 7. System Administration, Medical Staff & Integration Testing Flow

To support the clinic's scalability and enable full control without direct intervention in the code, the API exposes a set of administrative routes protected under the Role-Based Access Control (RBAC) scheme. This section documents the endpoints used for managing the directory of medical specialists and overall user administration, enabling audits, privilege escalation, and profile modifications.

Additionally, to certify the resilience of the backend development, the complete integration testing flow is documented. This flow simulates the behavior of an end user, sequencing HTTP requests from obtaining the access token, through the parametric query for specialists, to the execution and eventual cancellation of a medical appointment, validating that the server responds with the correct HTTP status codes at each step of the process.

**Medical Staff Module**

**(12) View All Doctors**

- HTTP Method: GET
- URI Endpoint: {{base_url}}/api/medicos
- Authorization Required: No
- Headers: None
- Payload: Without a body

Expected Response (200 OK): [ { "id": 1, "userId": 5, "fullName": "Dr. Carlos Mendoza Silva", "cedula": "1111111111", "email": "carlos.mendoza@flova.com", "telefono": "0991111111", "especialidad": "Fisioterapia", "numeroLicencia": "MED-FIS-001", "calificacion": 4.8, "pacientesAtendidos": 245, "horarioAtencion": [ { "diaSemana": 1, "horaInicio": "08:00", "horaFin": "16:00" }, { "diaSemana": 2, "horaInicio": "08:00", "horaFin": "16:00" }, { "diaSemana": 3, "horaInicio": "08:00", "horaFin": "16:00" }, { "diaSemana": 4, "horaInicio": "08:00", "horaFin": "16:00" }, { "diaSemana": 5, "horaInicio": "08:00", "horaFin": "16:00" } ] } ]

**(13) Get Doctor by ID**

- HTTP Method: GET
- URI Endpoint: {{base_url}}/api/medicos/1
- Authorization Required: No
- Headers: None
- Payload: Without a body

Expected Response (200 OK): { "id": 1, "userId": 5, "fullName": "Dr. Carlos Mendoza Silva", "cedula": "1111111111", "email": "carlos.mendoza@flova.com", "telefono": "0991111111", "especialidad": "Fisioterapia", "numeroLicencia": "MED-FIS-001", "calificacion": 4.8, "pacientesAtendidos": 245 }

**(14) Get Doctors by Specialty**

- HTTP Method: GET
- URI Endpoint: {{base_url}}/api/medicos/especialidad/Fisioterapia
- Authorization Required: No
- Headers: None
- Payload: Without a body

System Specialties Available: Fisioterapia, Terapia Ocupacional, Psicología

Expected Response (200 OK): [ { "id": 1, "userId": 5, "fullName": "Dr. Carlos Mendoza Silva", "cedula": "1111111111", "email": "carlos.mendoza@flova.com", "telefono": "0991111111", "especialidad": "Fisioterapia", "numeroLicencia": "MED-FIS-001", "calificacion": 4.8, "pacientesAtendidos": 245 } ]

**User Profiles Module**

**(21) Get All Users (Admin)**

- HTTP Method: GET
- URI Endpoint: {{base_url}}/api/users
- Authorization Required: Yes (Admin role required)
- Headers: Authorization: Bearer {{token_admin}}
- Optional Query Parameters: role (Allowed filters: paciente, medico, admin)
- Parametric Routing Example: {{base_url}}/api/users?role=medico

Expected Response (200 OK): [ { "id": 1, "cedula": "admin", "username": "admin", "fullName": "Administrador del Sistema", "email": "admin@flova.com", "telefono": "0999999999", "tipoSeguro": "ninguno", "role": "admin", "createdAt": "2026-05-02T03:43:01.000Z", "updatedAt": "2026-05-02T03:43:01.000Z", "medico": null }, { "id": 2, "cedula": "1234567890", "username": "1234567890", "fullName": "Juan Pérez García", "email": "juan.perez@email.com", "telefono": "0987654321", "tipoSeguro": "iess", "role": "paciente", "createdAt": "2026-05-02T03:43:01.000Z", "updatedAt": "2026-05-02T03:43:01.000Z", "medico": null } ]

**(22) Get User by ID**

- HTTP Method: GET
- URI Endpoint: {{base_url}}/api/users/2
- Authorization Required: Yes (User identity ownership or Admin

privilege)
- Headers: Authorization: Bearer {{token_paciente}}
- Payload: Without a body

Note: A normal user is authorized to request their unique profile details exclusively, whereas administrative roles are granted global profile search capabilities.

Expected Response (200 OK): { "id": 2, "cedula": "1234567890", "username": "1234567890", "fullName": "Juan Pérez García", "email": "juan.perez@email.com", "telefono": "0987654321", "tipoSeguro": "iess", "role": "paciente", "createdAt": "2026-05-02T03:43:01.000Z", "updatedAt": "2026-05-02T03:43:01.000Z", "medico": null }

**(23) Update User**

- HTTP Method: PUT
- URI Endpoint: {{base_url}}/api/users/2
- Authorization Required: Yes (User identity ownership or Admin privilege)
- Headers: Authorization: Bearer {{token_paciente}}, Content-Type: application/json

Body Variant A (Standard User Profile Update): { "fullName": "Juan Pérez García Actualizado", "email": "juan.nuevo@email.com", "telefono": "0987654399", "tipoSeguro": "privado" }

Body Variant B (Administrative Role Modification): { "role": "medico" }

Body Variant C (Credential Modification): { "password": "nueva_password_segura_123" }

Expected Response (200 OK): { "message": "Usuario actualizado exitosamente", "user": { "id": 2, "cedula": "1234567890", "username": "1234567890", "fullName": "Juan Pérez García Actualizado", "email": "juan.nuevo@email.com", "telefono": "0987654399", "tipoSeguro": "privado", "role": "paciente", "updatedAt": "2026-05-02T04:20:00.000Z" } }

**(24) Delete User (Admin)**

- HTTP Method: DELETE
- URI Endpoint: {{base_url}}/api/users/9

- Authorization Required: Yes (Admin role required)
- Headers: Authorization: Bearer {{token_admin}}
- Payload: Without a body

Expected Response (200 OK): { "message": "Usuario eliminado exitosamente" }

**Technical Notes and System Constraints**

**Required Headers**

Protected system endpoints mandate the execution of the following operational HTTP header settings: Authorization: Bearer <token_string> Content-Type: application/json

**Data Format Definitions**

- Calendar Dates: Handled as strict ISO strings using the YYYY-MM-DD pattern (Example: 2026-05-10).
- Time Windows: Handled using 24-hour notation strings in the HH:mm pattern (Example: 10:00).
- Identification Numbers (Cédula): Consists of a 10-digit numerical sequence or matches the root application string "admin".

**Valid Domain Enumerations**

- Insurance Providers: ninguno, iess, ejercito, policia, issfa, isspol, privado
- Medical Appointment Statuses: pendiente (created), confirmada (approved), completada (finished), cancelada (revoked)
- Application System Roles: paciente, medico, admin
- Medical Specializations: Fisioterapia, Terapia Ocupacional, Psicología

**Full Integration Testing Flow**

To evaluate the clinical system ecosystem from end to end, execute the following transactional steps in sequence:

**Step 1: User Session Authentication**

Verify client credentials against the infrastructure directory to receive the authorization token payload.

- Endpoint: POST {{base_url}}/api/auth/login
- Payload: { "cedula": "1234567890", "password": "password123" }

- Action: Retain and assign the returned "token" value for sub-routine authorization.

**Step 2: Therapeutic Catalog Inquiry**

Fetch the global dictionary of therapies to view treatment options.

- Endpoint: GET {{base_url}}/api/terapias

**Step 3: Medical Staff Filtering**

Query the clinical staff directory filtered by the target therapeutic specialty field.

- Endpoint: GET {{base_url}}/api/medicos/especialidad/Fisioterapia

**Step 4: Schedule Matrix Assessment**

Evaluate real-time open time-slots for the selected doctor to check for overlapping appointments.

- Endpoint: GET {{base_url}}/api/citas/horarios-disponibles?medicoId=1&fecha=2026-05-10
- Headers: Authorization: Bearer {{token}}

**Step 5: Transactional Resource Booking**

Generate a new clinical appointment entry containing the verified parametric choices.

- Endpoint: POST {{base_url}}/api/citas
- Headers: Authorization: Bearer {{token}}
- Payload: { "medicoId": 1, "terapiaId": 1, "fecha": "2026-05-10", "hora": "10:00", "sintomas": "Dolor en rodilla derecha después de correr.", "tieneExamenes": false }

**Step 6: User Record Verification**

Query individual appointment arrays to verify the successful persistence of the entry.

- Endpoint: GET {{base_url}}/api/citas
- Headers: Authorization: Bearer {{token}}

### Step 7: Booking Cancellation Flow

Initiate a soft-delete cancellation check to free the locked time-slot.

7. Endpoint: DELETE {{base_url}}/api/citas/1
8. Headers: Authorization: Bearer {{token}}
9. Payload: { "motivo": "Compromiso urgente de trabajo" }